Ultrasonic Atomization of Liquids

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Ultrasonic Atomization of Liquids

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(Received March 13, 1961)

An experimental study was made of the mechanism by which the ultrasonic vibration of liquid surfaces causes atomization.

At exciting frequencies in the range of 10 to 800 kc, uniform patterns of crossed capillary waves were found on the liquid surface when atomization occurred. The number-median diameter of the particles produced was found to be a constant fraction, 0.34, of the capillary wavelength; the capillary wavelength is calculable by Kelvin's equation using the exciting frequency and properties of the fluid being atomized.

The evidence is strong that the mechanism of ultrasonic atomization involves the rupture of capillary surface waves and the subsequent ejection of the wave peaks from the surface as particles.

I. INTRODUCTION

WHEN a beam of ultrasound of sufficient intensity is passed through a liquid and directed at an air interface, atomization of the liquid occurs. Liquid particles are ejected from the surface into the surrounding air. Under the proper conditions, very fine dense fogs are produced.

The technique of ultrasonic atomization has at least one advantage over conventional atomization techniques which makes it worthy of study; the fog particle size and fog density can be independently controlled. This cannot be done with other commonly used techniques for fine atomization. In pneumatic atomization, for example, particle size can only be reduced at the expense of fog density, because the air flow must be increased. In ultrasonic atomization the fog density can be varied simply by adjusting the air flow past the liquid surface. The amount of liquid which can be suspended in the air is limited only by the rate at which it falls out. The size of the particles in the fog can be varied by varying the frequency of the ultrasound; the higher the frequency the smaller are the droplets produced.

Not very much work has been published on the subject of ultrasonic atomization. The first description of this phenomenon was by Wood and Loomis,[1] who observed fog formation from an ultrasonic fountain and also off thin liquid films on a vibrating glass bulb. Söllner[2] investigated the mechanism of the atomization and concluded cavitation plays a major role. McCubbin[3] measured the size of the particles in a fog from a 2.4-Mc ultrasonic fountain and found them to be in the range of 1 to 10 μ. More recently, Bisa, Dirnagl, and Esche,[4] working at frequencies of 1.2 to 5.4 Mc, showed that particle size is a function of frequency, and suggested that capillary surface waves are involved in the atomization mechanism. They suggested further that particle size might be directly related to the wavelength of these capillary waves. Antonevich[5] measured the particle sizes produced at a low frequency of 20 kc. He also suggested that the rupture of capillary waves might be the prime cause of atomization.

The purpose of this present work was to follow out these suggestions: to determine if capillary waves do play a major role in the process of ultrasonic atomization and, if so, to find if a consistent relationship exists between the capillary wavelength at a given frequency and the size of the particles produced. This work covered a wide frequency range of 10 to 800 kc, intermediate between the ranges covered by previous investigators.

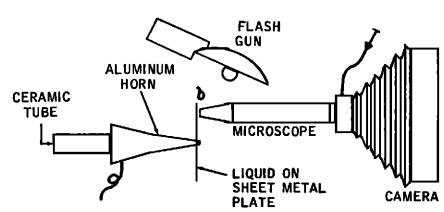


FIG. 1. Equipment for studying liquid surfaces vibrated at 10 to 20 kc.

II. EXPERIMENTAL TECHNIQUES

The experimental approach in this study was to measure the size of the disturbances caused by the ultrasound on the liquid surface for comparison with the measured particle size.

Several different techniques were used to activate the liquid surfaces, depending on the frequency. In all cases, it was necessary to concentrate the ultrasonic energy from the piezoelectric transducer.

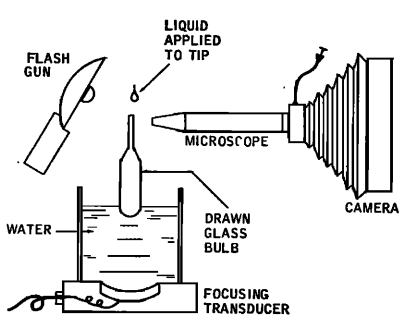


FIG. 2. Equipment for studying liquid surfaces vibrated at 400 to 800 kc.

[1] W. R. Wood and A. L. Loomis, Phil. Mag. 7, 417 (1927).
[2] K. Söllner, Trans. Faraday Soc. 32, 1532 (1936).
[3] T. K. McCubbin Jr., J. Acoust. Soc. Am. 25, 1013 (1953).
[4] K. Bisa, K. Dirnagl, and R. Esche, Siemens-Z. 28, 341 (1954).
[5] J. N. Antonevich, IRE Trans. on Ultrasonics Eng. PGUE-7, 6 (1959).

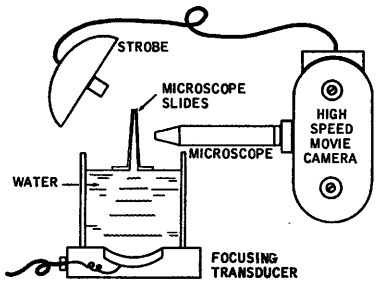


FIG. 3. Equipment for studying surface disturbances on a deep liquid.

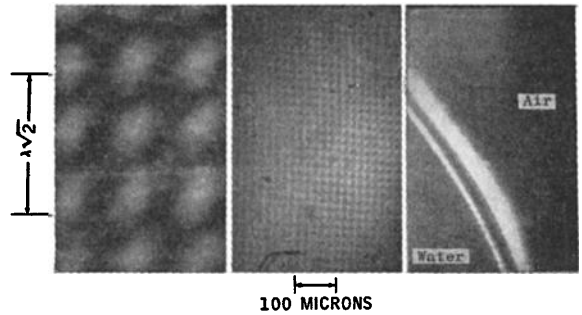


FIG. 4. Typical photographs of surface disturbances on water excited at various sound frequencies. Left: Thin film at 12 kc; Middle: Thin film at 390 kc; Right: deep liquid at 390 kc (edge view).

Most of the work was done with thin liquid films on vibrating solid surfaces. For low frequencies in the neighborhood of 10 to 20 kc, a half-wave tapered aluminum horn was used to vibrate a thin sheet metal plate in flexure as shown in Fig. 1. For intermediate frequencies of 100 to 400 kc, a drawn-out glass bulb was used. It was vibrated with focused sound from a concave reflector. To obtain high-frequency vibrations in the range of 400 to 800 kc, the glass bulb was activated with a bowl-shaped focusing transducer. A sketch of this equipment is shown in Fig. 2.

One experiment was made at high frequency using a deep pool of liquid. It was done to determine if the same type of surface disturbances occurred if the liquid depth was very great. The bowl-shaped focusing transducer was used for this work. The surface of the liquid jet formed by the sound radiation pressure was observed edgewise. This was done by "cutting a slice" out of this ultrasonic fountain with two closely spaced glass plates as shown in Fig. 3.

In all cases the dimensions of the surface disturbances were measured by photographing them. Water was the liquid used for most of the measurements, but the behavior of oil was also checked. The pictures were taken through a calibrated microscope while the surface was irradiated at a sound intensity slightly lower than that required for atomization. Higher sound levels were used to obtain atomization for particle size measurements.

Particle size was measured by atomizing molten wax. The wax droplets solidified quickly in flight, were collected, and sized. The particles produced at low frequencies were large enough that they could be sized by sieving. Another technique had to be used for the fine particles produced at the higher frequencies. The fine wax fogs were trapped in a glass cylinder and allowed to settle out on microscope slides. Photomicrographs were made of the slides and the particles were counted and sized manually.

Two different types of wax were used in these experiments. For sieving, a high-melting-point, nonsticky, synthetic wax was used to eliminate the clumping of particles. For the settling technique an ordinary paraffin wax was used because its lower melting point made it easier to handle. As shown in Table I, the oil used for taking some of the photographs of surface disturbances had physical properties similar to those of the molten waxes. The wax particles, therefore, should be very nearly the same size as particles would be if oil were atomized instead.

TABLE I. Physical properties of fluids atomized.

Liquid properties	Synthetic wax[a]	Paraffin wax	Oil
Temperature, °F[b]	310	133	100
Specific gravity/60°F	0.82	0.89	0.85
Surface tension, d/cm	21.7	30.0	27.0

[a] Acrawax C, manufactured by Glyco Products Inc.
[b] The wax temperatures given are slightly above the melting point and approximately those used for the atomization experiments.

III. RESULTS

The photomicrographs of the vibrating surfaces, of both water and oil, showed that the disturbances on the liquid surface were uniform patterns of crossed capillary standing waves. These waves were found at low and high excitation frequencies, on thin liquid films and when the liquid was deep. Typical photographs, taken perpendicular to a water surface, are shown in Fig. 4. A photograph taken parallel to the surface by the "slice" technique is also shown.

The outstanding difference in the appearance of the vibrating surfaces at various frequencies is in the wave-

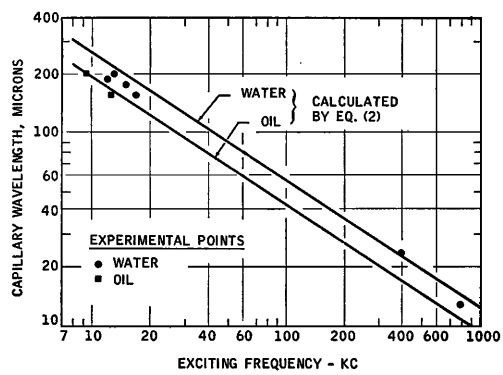


FIG. 5. Comparison of measured capillary wavelengths with those calculated for various exciting sound frequencies.

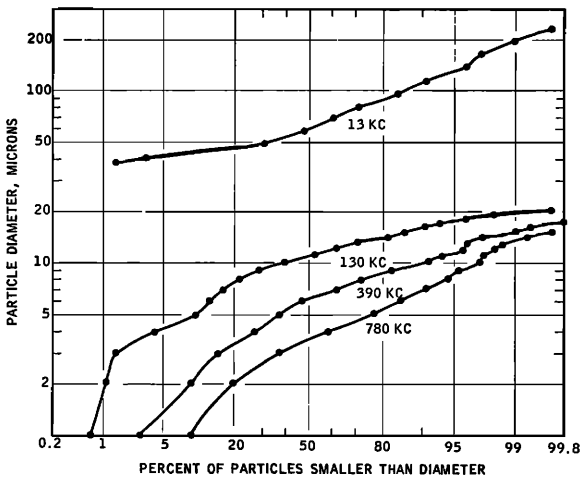


FIG. 6. Distributions of particle sizes in atomized wax samples.

length of the capillary waves. The waves were found to be finer at higher exciting frequencies. This is to be expected according to Kelvin's[6] equation for the wavelength of capillary waves:

$$\lambda^3 = 2\pi T/\rho f^2. \tag{1}$$

In Eq. (1), λ is the wavelength, T is surface tension, ρ is the liquid density, and f is the frequency of the waves on the surface of the liquid.

For sonically generated capillary waves, the surface wave frequency in Eq. (1) has been shown to be equal to one half the exciting sound frequency.[7] In terms of the sound frequency, the capillary wavelength is

$$\lambda = (8\pi T/\rho F^2)^{\frac{1}{3}}, \tag{2}$$

where F = exciting sound frequency in cps.

Measurements of the capillary wavelengths taken off the surface photographs are plotted in Fig. 5 for two liquids, oil and water. They agree very well with the results predicted by Eq. (2).

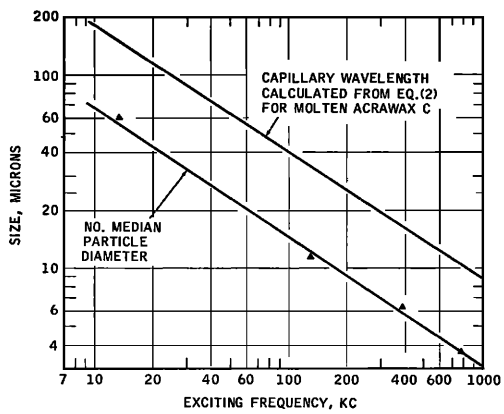


FIG. 7. Particle size and calculated capillary wavelength for various exciting sound frequencies.

Particle-size data obtained in these experiments are plotted in Fig. 6. The particles produced at a given frequency were found not to be uniform in size. The distributions of particle diameters in the samples were such that roughly 90% of the particles were smaller than twice the number-median diameter.

The number-median particle size is compared with capillary wavelength in Fig. 7. The data show that the particle size is a very nearly constant fraction of the capillary wavelength. A good correlation is obtained with a modification of Eq. (2):

$$D = 0.34(8\pi T/\rho F^2)^{\frac{1}{3}}, \tag{3}$$

where D = number median particle diameter in cm.

It is important to state here that the samples of particles taken in this study were produced at very low atomization rates of about 0.01 cc/sec. Higher atomization rates tended to produce larger particles.

IV. DISCUSSION

On the basis of these data, it appears that capillary waves do play a major role in the process of ultrasonic atomization.

The mechanism by which the capillary waves give rise to atomization probably is as follows. Capillary wave peaks, under a sufficiently strong exciting vibration, build up in amplitude until they become unstable. At this point, the wave peaks break loose from the bulk of the liquid and are ejected from the surface into the air as droplets.

The median particle diameter produced by ultrasonic atomization, at low liquid feed rates, is predictable. It is 0.34 times the capillary wavelength calculated from Kelvin's equation, presented in this paper as Eq. (2).

It is reasonable, according to the envisioned atomization mechanism, that the median particle diameter should be less than half of a capillary wavelength, for only a part of the wave peak is pictured as being ejected from the surface. The fact that the particles produced are not uniform in size may mean there is some variability in how much of the wave peak tears loose. On the other hand, the nonuniformity could be caused by collision and agglomeration of the droplets shortly after leaving the surface. This latter position finds support in the fact that particle size increases with atomization rate.

These data tend to confirm the suggestions of previous workers,[4,5] that capillary surface waves give rise to ultrasonic atomization. There is need, however, for more work to be done before the detailed mechanisms involved are known.

[6] J. W. S. Rayleigh, *The Theory of Sound* (Dover Publications, New York, 1945), Vol. II, Chap. 20, p. 344.
[7] W. Eisenmenger, Acustica 9, 327 (1959).